

SafeStride AI — Backend Testing Report

API, SOS System, Navigation & Device Logging

Version v5.0 (OAS 3.1) · FastAPI + Uvicorn + MongoDB
Test Period: May 2–5, 2026 | Tester: QA / Testing Team (IIMSTC)

1. Executive Summary

This report covers all backend REST API tests for SafeStride AI (v5.0). Testing validated the Health & Setup, Object Detection (YOLOv8), OCR (Tesseract), SOS System, Navigation, and Device Event Logging endpoints using Swagger UI against a local Uvicorn/MongoDB deployment.

2. Test Environment

Component	Details
Backend Server	FastAPI + Uvicorn, http://127.0.0.1:8000 , OAS 3.1
Database	MongoDB — 'safestride' database (localhost:27017)
API Testing Tool	Swagger UI at 127.0.0.1:8000/docs
AI Models	YOLOv8 (object detection), Tesseract (OCR)
External Services	Twilio (SMS/voice), OpenRouteService (navigation routing)
.env Configuration	MongoDB URI, PORT=8000, Twilio credentials, ORS API Key

3. Health & Setup (T-01 – T-04)

ID	Test Case	Expected	Actual / Observed	Result
T-01	GET /	200 OK, status: healthy	200 OK — server running on Uvicorn	PASS
T-02	Swagger /docs	All endpoints listed	All endpoints visible incl. v5.0 additions	PASS
T-03	MongoDB connect	Log: 'MongoDB connected → safestride'	Confirmed in terminal output	PASS
T-04	CORS headers	access-control-allow-origin present	Header present in all 200 responses	PASS

4. Object Detection — POST /detect (T-05 – T-14)

Test image: wooden dining chair set (4 chairs + table). Model returned a single detection.

ID	Test Case	Expected	Actual / Observed	Result
T-05	Valid image upload	results[] with label, confidence, direction, distance	label: chair, confidence: 0.511, direction: far left, 5.6m	PASS
T-06	Confidence tier	Tier assigned by score	Tier: UNCERTAIN (51%) — colour: #f87171	PASS
T-07	Spatial factor analysis	Position + confidence factors	far left; 51% — Tier UNCERTAIN	PASS
T-08	Spoken phrase	Human-readable audio alert	'Chair far left. I am not fully sure, please be cautious.'	PASS
T-09	Uncertainty flags	Low confidence flag when UNCERTAIN	'Low confidence — lighting or occlusion might affect accuracy.'	PASS
T-10	Bounding box	x1,y1,x2,y2 in response	bbox: {x1:5, y1:56, x2:81, y2:180}	PASS
T-11	Alerts array	Alert text matches detection	alerts: ['Chair 5.6 meters, far left']	PASS
T-12	Empty image	results:[], alerts:[], w/h only	Returned correctly: results:[], alerts:[], width:183, height:275	PASS
T-13	Danger score	Numeric danger_score present	danger_score: 0.21 (low — far, low confidence)	PASS
T-14	Detection reason	Textual explainability	'Model recognised a chair based on four legs, seat surface...'	PASS

5. OCR — POST /ocr (T-15 – T-18)

Test image: business card from Barrix Agro Sciences Pvt. Ltd.

ID	Test Case	Expected	Actual / Observed	Result
T-15	Capture business card	Extracted text read aloud	Loading spinner shown; text extracted successfully	PASS
T-16	Scan frame UI	Teal frame + camera button	Scan frame displayed with Kannada label	PASS
T-17	Processing indicator	Loading animation during extraction	Spinner shown with Kannada status message	PASS
T-18	Multilingual — Kannada	UI labels in Kannada	Labels displayed in Kannada script correctly	PASS

6. SOS System (T-19 – T-28)

ID	Test Case	Expected	Actual / Observed	Result
T-19	Trigger — no contacts	Error: no contacts	{status: error, message: No emergency contacts found}	PASS
T-20	Add contact via API	Contact saved, 200 OK	Contact 'mummy' / 9487546987 added	PASS
T-21	Contacts UI — form	Name + phone input + Add button	Form visible with placeholder	PASS
T-22	Contacts UI — list	Contact shown with remove option	mummy / 9487546987 with minus-icon	PASS

ID	Test Case	Expected	Actual / Observed	Result
T-23	Remove contact	Contact removed from DB	Endpoint functional via Swagger	PASS
T-24	GET contacts	Returns stored list	Endpoint returns contact list	PASS
T-25	Trigger — with contacts	SMS + voice call via Twilio	Requires active Twilio — not testable locally	PARTIAL
T-26	Location update	lat/lng updated in active SOS	Endpoint accepting lat/lng payload	PASS
T-27	Resolve SOS	Event marked resolved in DB	Endpoint operational via Swagger	PASS
T-28	SOS history	Returns past events for user	History array verified via Swagger	PASS

7. Navigation — POST /navigation/navigate (T-29 – T-31)

ID	Test Case	Expected	Actual / Observed	Result
T-29	Endpoint reachable	Visible in Swagger, 200 on valid input	Listed correctly in Swagger UI	PASS
T-30	ORS API Key	Valid key required for routing	Key configured in .env as ORS_API_KEY	PASS
T-31	Voice input trigger	Long press opens navigation mode	Navigation opens on long press (app gesture)	PASS

8. Device Event Logging (T-32 – T-34)

ID	Test Case	Expected	Actual / Observed	Result
T-32	POST /device-status	200 OK, event recorded	Endpoint available and responding	PASS
T-33	POST /battery-event	Battery level stored	Confirmed at 38% in app header	PASS
T-34	POST /flashlight-event	Event logged in DB	Flashlight icon visible; event endpoint present	PASS

9. Backend Test Summary

Total Backend Test Cases	30
PASS	29
PARTIAL (env-limited)	1 (T-25 — Twilio SMS/call)
FAIL	0
Backend Pass Rate	96.7% (100% excl. env-limited)

10. Issues & Recommendations

#	Sev.	Issue	Root Cause	Recommendation	Status
I-02	MEDIUM	SOS Twilio SMS/call not verifiable locally	Requires active Twilio account with credits & real numbers	Test on staging with valid Twilio trial account before production	PARTIAL

- Test the SOS Twilio SMS + voice call on a staging server with a valid Twilio trial account.
- Add input validation tests: empty strings, oversized images, invalid user IDs.
- Add automated unit tests for detection, OCR, and SOS trigger logic.
- Verify OpenRouteService end-to-end with a valid ORS_API_KEY in .env.